Key-Based Enhancement of Data Encryption Standard For Text Security

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***Abstract*—Securing various data types such as text, image, and video is needed in real-time communications. The transmission of data over an insecure channel is a permanent challenge, especially in mass Internet applications. Preserving confidentiality and integrity of data toward malicious attacks, accidental devastation, change during transfer, or while in storage must be improved. Data Encryption Standard (DES) is considered as a symmetric-key algorithm that is most widely used for various security purposes. In this work, a Key-based Enhancement of the DES (KE-DES) technique for securing text is proposed. The KE-DES is implemented by the application of two steps: the first is merging the Odd/Even bit transformation of every key bit in the DES algorithm. The second step is replacing the right-side expansion of the original DES by using Key-Distribution (K-D) function. The K-D allocation consists of 8-bits from Permutation Choice-1 (PC-1) key outcome. The next 32-bits outcomes from the right-side of data, there is also 8-bits outcome from Permutation Choice-2 (PC-2) in each round. The key and data created randomly, in this case, provide adequate security and the KE-DES model is considered more efficient for text encryption.**

***Index Terms*—Data Encryption Standard, Text Security, CBC Mode, Key-Distribution Function**

I. Introduction

Cryptography and network security is a concept to protect the network and data during transmission over an unreliable network [1]. Data encryption is the main aspect of preserving the confidentiality and integrity of transferred information. This is regarded today as a challenging data transmission problem that affects many areas, including secure communication channels, strong data encryption techniques, and reliable database maintenance by third parties [2]. The traditional obfuscation techniques can only preserve data security. Cryptography is also a key technology for essential electronic systems. It is used to keep that information confidential, digitally verify transactions, access control, and so on. During cryptographic protocols, main terms are used such as plaintext, ciphertext, and the key-space [3]

Data Encryption Standard (DES) is considered as one of the most widely used symmetric encryption algorithms for different goals. Its main designing thought is still used in considerable block ciphers [4]. Symmetric methods use a series of characters and numbers and sometimes binary strings (bits) shifting operations to mutate the message. Since its presentation in November 1977 as the Federal Information Processing Standard (FIPS) [5], the DES algorithm has played a significant role in data security over the last four decades. Since then, various encryption methods for different data types such as text, image, and video has been suggested in literature [6]–[8].

Much concern has been raised concerning the protection of data transmitted or stored over open channels throughout the channel innovations of the communications industry. The main priority is about plaintext messages [9]. Every other cryptosystem aims to maintain the privacy of data in electronic distribution systems. Also, there is a need for encrypted transmission of confidential information. So that such an authoritative person can decipher it. Any algorithm's strength is based on a secret value or key-space size. Without knowledge of the key, breaking cryptographic algorithms is considered a dilemma [10].

II. Related Work

Since 1977, DES has played a crucial role in data security systems. Among the three key features of the security services (i.e. confidentiality, integrity, and availability), confidentiality is the most important aspect for the sharing of plaintext messages between users [11]. In [12], a change to the DES is suggested to make it safe and susceptible to bit errors caused by the wireless channel. It is noted that the use of the modified algorithm in these channels enhances the performance of the bit error rate as well as the security comparison with DES. For the security of data using smart cards, authors in [13] have developed an enhanced DES structure. The simulation of the project gives a useful initiation for exploring a far more robust encryption system that doesn't require so many arithmetic operations. A mixed DES and SHA-1 cryptographic system has been developed in [14] based on the Visual C++ environment. The suggested model is used 3DES and RSA data encryption algorithms for hiding and the SHA-1 algorithm to validate data integrity. An enhanced key generation unit is proposed in [15]. Using the proposed design, high speed and reduced logic complexity are achieved, resulting in an enhanced DES algorithm. The length of the key was

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increased by the authors in [16], which improved the DES algorithm. The suggested technique performed even better than the previous algorithm for detecting the encryption key or the total number of keys that the blind search method could generate. An enhanced DES used f-function by incorporating striding techniques rather than just conducting XOR operation between the 56-bit sub-key and the plaintext is proposed in [17]. This resulted in an average avalanche effect of 55% due to single-bit vibration in plaintext. The authors in [18] proposed an enhanced secure DSE (ES-DES) model using an extended substitution box (S-Box) to support and increase the security of the DES algorithm. The safety of the ES-DES method is improved against possible attacks and the confusion process makes it difficult to deducing the key. An enhanced DES (EDES) using variable size key and 96-bit sub-key is proposed in [19]. EDES can be used for communication with secret information and also for files over computer networks. By enhancing the key length and the limited round function toward cryptographic attacks, the authors in [20] proposed a modified DES model. Evidence has shown that the probability of a characteristic for differential cryptanalysis is reduced and the distance of unicity is better than the standard DES.

III. The Proposed Key-Based Enhanced Model

This section describes the presented KE-DES algorithm and the improvement overview done in this work. It also gives more details about key-space usage.

A. Description of KE-DES

The KE-DES algorithm is designed to encrypt and decrypt a 64-bit data block under 64-bit key control [21]. Deciphering must be achieved by using the same key as the enciphering side, but with a schedule of addressing key bits changed such that the deciphering process is the opposite of the enciphering process. In the KE-DES algorithm, the standard expansion function is not used directly but instead, a key-distribution (K-D) function is created. The new function consists of 8-bits from Permutation Choice-1 (PC-1) as input key resulted bits and it will be constant with all rounds. There is also 8-bits outcome from Permutation Choice-2 (PC-2) that is changing in each round will be considered. The next 32-bits outcomes from the right-side (R) in each round, so that the total value in the new expansion operation will be 48-bits as usual. The K-D process is depicted in Figure 1.

B. Key Generation in KE-DES

The suggested KE-DES model needs 64-bits as the input key. The following algorithm describes the steps of the key generation process that occur in this method:

1) Permuted Choice-1 (PC-1) will be used to create a key size of 56-bits as shown in Figure 2.
2) In every Even bit position, 1 and 0 will be replaced by Odd bit position in the 56-bits block. The pseudo-code computation is used as given in Listing 1:

Listing 1. **Odd/Even Operation Pseudo Code**

```
function Odd/Even (Binary B)
  For j = 1 to size of B
    If j Mod 2 is equal to 0 then
        B[j] = 1;
    else
        If j Mod 2 is equal to 1 then
            B[j] = 0
        end
    end
  end
end function
```

3) Divide the result into two halves of 28-bits (C_0 and D_0)
4) Left shifts are performed to the previous results as in Table I to obtain C_1 and D_1.
5) Apply PC-2 as in Figure 2 to obtain K_1 from concatenation of $K_1 = C_1||D_1$.
6) Concatenate C_1 and D_1 to achieve 56-bits block and use this as input to the next round to obtain C_2 and D_2, C_3 and D_3, and so forth.
7) Repeat steps 2 to 6 to obtain K_2 to K_{16} values and iterate the subscript value by one. This is done for all the 16 rounds as shown in Figure 4.

C. Encryption and Decryption by the KE-DES

The first step in ciphering 64-bits input data block is the initial permutation (IP), the second step is a complex key-dependent calculation and finally an inverse of the initial permutation IP^{-1} as presented in Figure 3. For our work, we will depend on divided the text file into blocks by using Cipher Block Chaining (CBC) mode [22] where each block consists of 64-bits and then, apply the proposed KE-DES encryption process as depicted in Figure 4.

The modified encryption process will be done by swapping the expansion permutation function using the created K-D distribution function. The main functionality of this procedure is described by the following steps:

1) Use the IP in Figure 3 for the input plaintext.
2) The permutation result is divided into two halves blocks: Left (L_0) and Right (R_0). Each half has a length of 32-bits.
3) Apply K-D operation to R_0.
4) Perform XOR operation of step 3 result with K_1.
5) Apply S-Box substitution operation for step 4 result.
6) Apply standard permutation (P) in Table II for step 5 result.
7) Compute the value of R_1 by XORing the result of step 6 with L_0.
8) Assign the value $L_1 = R_0$ to complete round no 1.
9) Repeat steps 3 to 8 for the next rounds to obtain L_{16} and R_{16}.
10) Finally, apply IP^{-1} as presented in Figure 3 to obtain the ciphertext.

The decryption process is the reverse process of encrypting plaintext data. In this step, the ciphertext is considered to be

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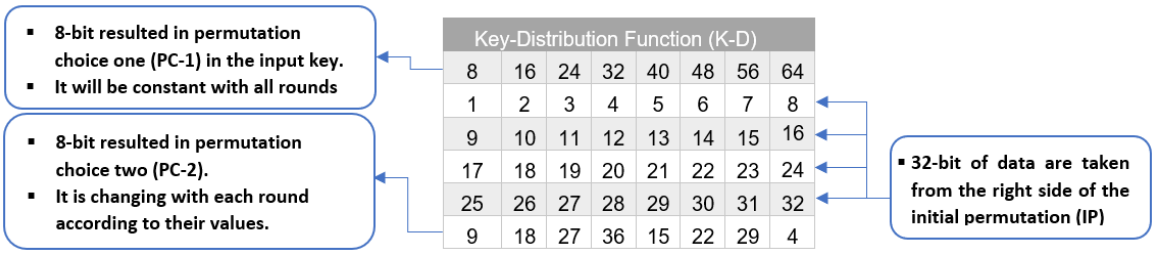


Fig. 1. Key and data distribution in the KE-DES model

Permuted Choice-1 (PC-1)						
57	49	41	33	25	17	9
1	58	50	42	34	26	18
10	2	59	51	43	35	27
19	11	3	60	52	44	36
63	55	47	39	31	23	15
7	62	54	46	38	30	22
14	6	61	53	45	37	29
21	13	5	28	20	12	4

Permuted Choice-2 (PC-2)					
14	17	11	24	1	5
3	28	15	6	21	10
23	19	12	4	26	8
16	7	27	20	13	2
41	52	31	37	47	55
30	40	51	45	33	48
44	49	39	56	34	53
46	42	50	36	29	32

Fig. 2. PC-1 and PC-2 in KE-DES method

TABLE I
LEFT SHIFT BOX

Shift Schedule (Left Shifts)																
Number of rounds	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Number of shifts	1	1	2	2	2	2	2	2	1	2	2	2	2	2	2	1

(a) Initial Permutation (IP)							
58	50	42	34	26	18	10	2
60	52	44	36	28	20	12	4
62	54	46	38	30	22	14	6
64	56	48	40	32	24	16	8
57	49	41	33	25	17	9	1
59	51	43	35	27	19	11	3
61	53	45	37	29	21	13	5
63	55	47	39	31	23	15	7

(b) Inverse Initial Permutation (IP^{-1})							
40	8	48	16	56	24	64	32
39	7	47	15	55	23	63	31
38	6	46	14	54	22	62	30
37	5	45	13	53	21	61	29
36	4	44	12	52	20	60	28
35	3	43	11	51	19	59	27
34	2	42	10	50	18	58	26
33	1	41	9	49	17	57	25

Fig. 3. Bits locations of IP and IP^{-1} functions

an input to the KE-DES algorithm structure for decryption. Then the same hidden key is the decryption key used for the encryption process. Here we obey the reverse order encryption protocol. After the decryption has been completed, the output obtained is known to be the plaintext, following the same characteristic of the original text.

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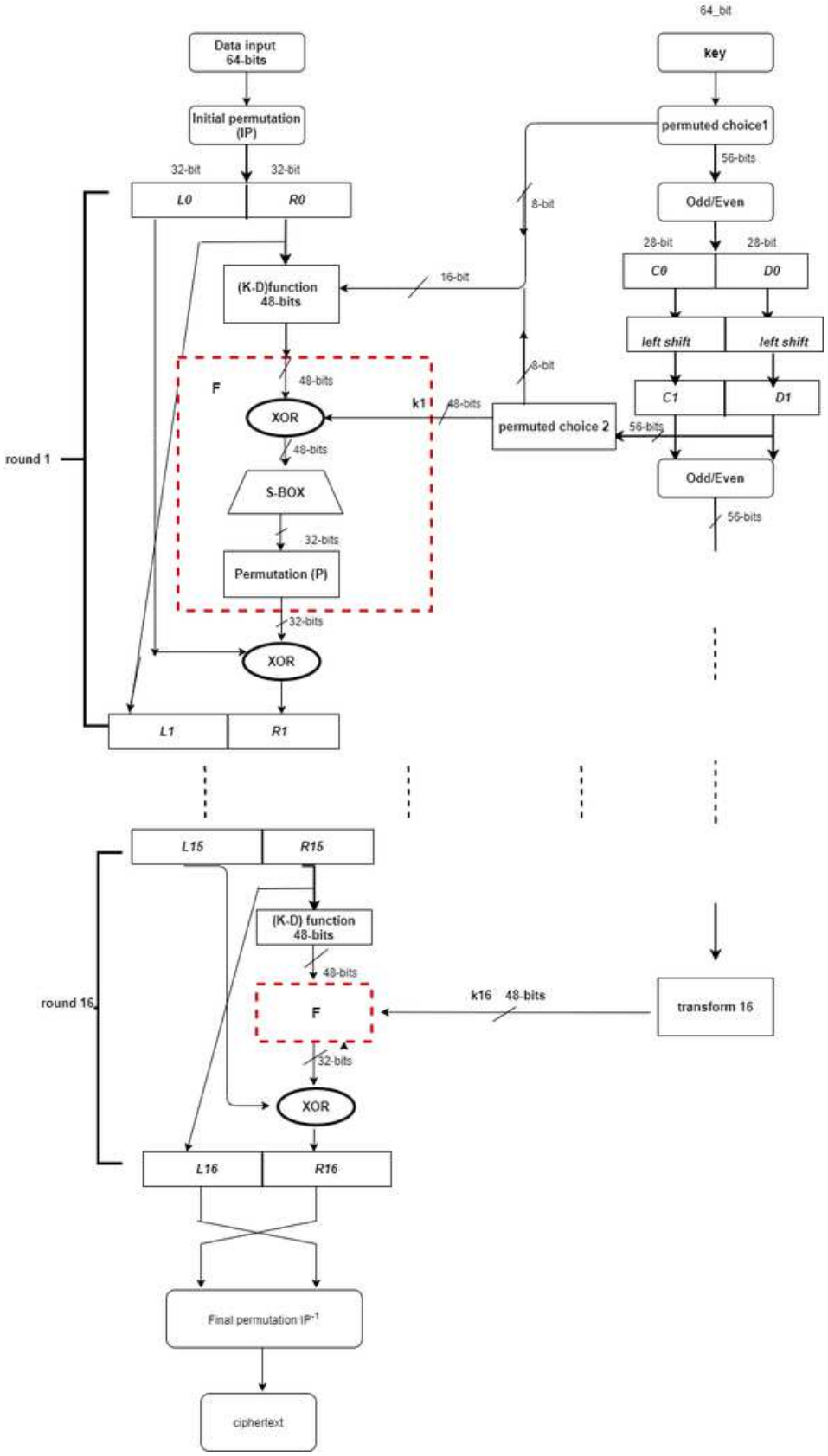


Fig. 4. Key generation and encryption process with KE-DES

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TABLE II
STANDARD PERMUTATION FUNCTION

Permutation Function (P)			
16	7	20	21
29	12	28	17
1	15	23	26
5	18	31	10
2	8	24	14
32	27	3	9
19	13	30	6
22	11	4	25

IV. RESULTS AND DISCUSSION

In this paper, the KE-DES encryption scheme was implemented in C++ programming software. Here is an example of using the KE-DES as a block cipher which operates on plaintext blocks of 64-bits size and produces ciphertext blocks of the same size. Our PC setup was: Probook Intel (R)Core i7-2670QM CPU @ 2.20 GHz, 8−GB RAM with64-bit Operating System. The file size of text data is 10-KB.

The first step is created 16-round keys. Let the key value be in hexadecimal = 133457799BBCDFF1

1) Convert key to 64-bits: 00010011 00110100 01010111 01111001 10011011 10111100 11011111 11110001.
2) The key after PC-1: 01111000 01100110 01010101 01111010 10101011 00110011 11000111.
3) Function Odd/Even result is: 01010101 01010101 01010101 01010101 01010101 01010101 01010101.
4) Divide the output into halves C_0 and D_0 of 28-bits each, then apply left shift box to obtain C_1 and D_1.
5) C_1 and D_1 are concatenated to fulfill 56-bits, then this is used as input to the next round.

TABLE III
C AND D VALUES FOR THE 16-ROUNDS

C_0	01010101010101010101010101
D_0	10101010101010101010101010
C_1	10101010101010101010101010
D_1	01010101010101010101010101
.....	.....
C_{16}	01010101010101010101010101
D_{16}	10101010101010101010101010

6) Apply PC-2 to Concatenate C_1 and D_1 and so on as listed in Table III to obtain the keys $K_1, K_2, ..., K_{16}$ as presented in Table IV

TABLE IV
ROUND KEY VALUES

R_key 1	011011101010110000011010010000110001100110111101
R_key 2	100100010101001111100101101111001110011001000010
.....	.....
R_key 15	011011101010110000011010010000110001100110111101
R_key 16	100100010101001111100101101111001110011001000010

The second step is divided input data into blocks and following the next two steps:

1) Apply the CBC mode to divide the plaintext file into 8-byte blocks represents 64-bits in binary system.
2) The plaintext is shown in Figure 5 (a) and the corresponding ciphertext is presented in Figure 5 (b).

Table V showed a comparison between the suggested KE-DES model and the original DES algorithm for one block of data. The key generation operating time, and the numbers of estimations of both the two methods are given. The finding results are promised for the usage of KE-DES method in various data types security applications.

V. CONCLUSIONS

In this work, an enhanced version of the DES algorithm was produced called the KE-DES. In this model, a new K-D function is created for the key and data to ensure greater security and more effectiveness in textual data encryption. Applying this model to other types of data could be interesting using reforming optimizations of the KE-DES algorithm in the future. Also, developing the model to the level of triple-KE-DES is currently in progress to expand the key-space usage and reduce the computation time.

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The Cipher Block Chaining (CBC) mode encrypts a preprocessed block of a message instead of an original message block, where the preprocessed block is formed through exclusive-ORing the original message block with a cipher data block produced from encrypting the previous message block. In the case of the first preprocessed block, a data block called initialization vector is used for exclusive Ring operation with the first message block. In other words, CBC encryption is a process of chaining successive cipher and message blocks together until the last message block is encrypted.

(a)

CB56922039AE9D6B40B0F68B26228259EE3366A2EC8C19E116DA84A34BA26FDE7CE603194F128E9C1FF576928267
6EDAA4F814A9C7768C54D94B689D2491A81409A17B3F431C6FE6366C9AB3AD12AF6620A4542AD0B85C9F36697C
8A0434DD8FD7A251B9127FC4E276085692B03C8B9A0BA600B240795AD8CEF00D08F5B07D5CCBBB03A1444B5DD
033323B0B250E1691F464C99734F45A8FE2226339E4EE431828EA7C21FE07F6D4E8C76E395E4556AC02A9CB2B8F3
AD3183780B3AE5E64E0A0A06587A22BBDE0EE7D21FDB6455479367EB0AC33B9B03991A91E96908E9F11779226A
C25D776935EA6403FB30F8CF79D11A02BAC7EDD9ED07E83E382B567D5E6A332FDB995EF98176EE52AB96660B55
11AF9B9931A9610D69951520ACEB89A9D48A264235DAB34E583ABAF03FE5C0CDF4C25F0309839B8BF6F331D963
0D2AD3E2AD146B106874F51D0D22B010D0F6A1C90A1E08E78B65F9D4E2C329975807CAB5832D12203C3641A4962
DE8C05EB6E1ED429AD115E752306B362DC27E68B26DE51876FBC0C6CB07F0DE2C0AF71BE69A32092C6B1C0124
EA88ECBED998D496FD9DC5A0D0659AB575A85593B220FB32B94A366D2A458CC17487AE3EA801D0743D57FDDA
9B8677A27F76FD3F6282D910E532C1A83858C598A571553B80205295CB544A59499E02B7C525EA8CBAE01FEDF5C
DE9139A6A43A502F6FF93D980C566452879D5EB6EED21F51CB65A30976B052DF84908FA55A4C591B2EE77BBB42
B985B46C303DA24BB11346F6594D9F2FB71A1D4BA369A7A7A30E676F5D8183190A97B7F7531B2835E09535B36386
03823DA509

(b)

Fig. 5. Plaintext encryption example and the produced ciphertext

TABLE V
COMPARISON OF THE NUMBER OF ESTIMATIONS FOR THE TWO ALGORITHMS WITHIN ONE DATA BLOCK

Parameters	PC-1	PC-2	The offset and the number of times the key is exchanged	Key generation Ttime (ms)
DES	1	16	16	0.001
KE-DES	1	16	16	0.002

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